CMR INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
Principles of Communication Systems
Multiple choice questions
(Answers are highlighted in red)

1. The number of voice channels that can be accommodated for transmission in T1 carrier system is
   a. 24
   b. 32
   c. 56
   d. 64

2. The technique that may be used to reduce the side band power is
   a. MSK
   b. BPSK
   c. Gaussian minimum shift keying
   d. BFSK

3. In uniform quantization process
   a. The step size remains same
   b. Step size varies according to the values of the input signal
   c. The quantizer has linear characteristics
   d. Both a and c are correct

4. The process of converting the analog sample into discrete form is called
   a. Modulation
   b. Multiplexing
   c. Quantization
   d. Sampling

5. The characteristics of compressor in μ-law companding are
   a. Continuous in nature
   b. Logarithmic in nature
   c. Linear in nature
   d. Discrete in nature

6. The modulation techniques used to convert analog signal into digital signal are
   a. Pulse code modulation
   b. Delta modulation
   c. Adaptive delta modulation
   d. All of the above

7. The sequence of operations in which PCM is done is
   a. Sampling, quantizing, encoding
   b. Quantizing, encoding, sampling
   c. Quantizing, sampling, encoding
   d. None of the above

8. In PCM, the parameter varied in accordance with the amplitude of the modulating signal is

a. Amplitude
b. Frequency
c. Phase
d. None of the above

9. One of the disadvantages of PCM is
a. It requires large bandwidth
b. Very high noise
c. Cannot be decoded easily
d. All of the above

10. In Delta modulation,
a. One bit per sample is transmitted
b. All the coded bits used for sampling are transmitted
c. The step size is fixed
d. Both a and c are correct

11. In digital transmission, the modulation technique that requires minimum bandwidth is
a. Delta modulation
b. PCM
c. DPCM
d. PAM

12. In Differential Pulse Code Modulation techniques, the decoding is performed by
a. Accumulator
b. Sampler
c. PLL
d. Quantizer

13. DPCM is a technique
a. To convert analog signal into digital signal
b. Where difference between successive samples of the analog signals are encoded into n-bit data streams
c. Where digital codes are the quantized values of the predicted value
d. All of the above

14. DPCM suffers from
a. Slope over load distortion
b. Quantization noise
c. Both a & b
d. None of the above

15. The noise that affects PCM
a. Transmission noise
b. Quantizing noise
c. Transit noise
d. Both a and b are correct

16. The factors that cause quantizing error in delta modulation are

a. Slope overload distortion
b. Granular noise
c. White noise
d. Both a and b are correct

17. Granular noise occurs when
a. Step size is too small
b. Step size is too large
c. There is interference from the adjacent channel
d. Bandwidth is too large

18. The digital modulation technique in which the step size is varied according to the variation in the slope of the input is called
a. Delta modulation
b. PCM
c. Adaptive delta modulation
d. PAM

19. In Adaptive Delta Modulation, the slope error reduces and
a. Quantization error decreases
b. **Quantization error increases**
c. Quantization error remains same
d. None of the above

20. T1 carrier system is used
a. For PCM voice transmission
b. For delta modulation
c. For frequency modulated signals
d. None of the above

21. Matched filter may be optimally used only for
a. Gaussian noise
b. Transit time noise
c. Flicker
d. All of the above

22. Characteristics of Matched filter are
a. Matched filter is used to maximize Signal to noise ratio even for non Gaussian noise
b. It gives the output as signal energy in the absence of noise
c. They are used for signal detection
d. All of the above

23. Matched filters may be used
a. To estimate the frequency of the received signal
b. In parameter estimation problems
c. To estimate the distance of the object

d. All of the above